

Memoria

Developing Open, Interoperable Semantic Resources

Grupo 6

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Introducción

En este proyecto se ha desarrollado una ontología centrada en el dominio del cine. El objetivo principal es representar de forma estructurada información relacionada con películas, actores, directores y otros elementos del ámbito del cine.

Todo se ha desarrollado a partir del dataset inicial “U.S. movies with gender-disambiguated actors, directors, and producers” que se encuentra en el repositorio del grupo.

Además de la ontología, también se han generado SHACL shapes para validar los datos y un grafo de conocimiento que sigue el modelo definido. Durante el desarrollo se reutilizaron recursos semánticos existentes y se utilizaron distintas herramientas para evaluar la ontología, siguiendo principios FAIR.

Desarrollo

Análisis de los datos

Nuestros datos muestran información relacionada con películas, actores, productores y directores. Hemos complementado estos datos con otros datasets que estudian los premios Oscar y las puntuaciones recibidas en IMDB. Los datos están enfocados con una clara intención de estudio de género en este ámbito por el recuento de pronombres masculinos y femeninos para cada película, por ejemplo. En los siguientes apartados exponemos en más detalle la información presentada por estos datasets.

US movies with gender disambiguated actors, directors and producers

Los datos que tenemos en el dataset de “DOISR-Core” describen Directores, Actores, Productores y Películas.

Para Directores se describen las siguientes características:

- ID
- Nombre
- Género (Masculino o Femenino)
- Primera y última película
- Lista de películas
- Rol (casting primario, casting secundario y director)
- Recuento de pronombres por género (número de Hombres y número de mujeres)
- Indicadores de créditos de actores (créditos de actores y de actrices)

Para Actores:

- ID
- Nombre
- Género (Masculino o Femenino)
- Lista de películas

Para Productores:

- ID
- Nombre
- Género (Masculino o Femenino)
- Primera y última película
- Lista de películas
- Recuento de pronombres por género (número de Hombres y número de mujeres)
- Indicadores de créditos de actores (créditos de actores y de actrices)

Y, por último, las películas:

- ID
- Título
- Año de estreno
- Género
- Presupuesto (ajustado por inflación)
- Información del reparto (lista de actores y actrices)
- Información de directores (lista de directores (suele ser 1))
- Información de productores (lista de productores)

IMDB Ratings

El dataset tiene 67408 filas y 5 columnas.

Tiene una columna índice que se descartará en el preproceso de los datos, un ID único para el título, un título, la puntuación recibida y el número de votos que han contribuido a esa puntuación.

Oscar Nominees and winners (1929-present)

El dataset tiene 11002 filas y 10 columnas y contiene información sobre los premios Óscar de películas desde 1929 hasta el presente.

Las columnas son:

- Year_film: El año de estreno de la película
- Decade_film: Década de estreno de la película
- Release_date: Fecha de estreno de la película
- Year_ceremony: Año en el que tuvo lugar la ceremonia de los Óscar
- Ceremony: Número de ceremonia
- Film: Título de la película usado en la ceremonia
- Original_title: Título original de la película
- Name: Persona o entidad nominada (Actor, Director, Productor, etc.)
- Category: Categoría de los premios Óscar
- Ganador: Booleano que dice si se ganó o no

Requisitos

- R1;
Competency question: What movies are represented in the dataset?
Answer: Movies from IMDb
Status: Proposed
Provenance: movies.json
Relation with other req: Root requirement for movies
Associated ontology Req: Movie
- R2;
Competency question:
What actors are represented in the dataset?
Answer: Actors from IMDb
Status: Proposed
Provenance: actors.json
Relation with other req: Root requirement for actors
Associated ontology Req: Actor
- R3;
Competency question:
What directors are represented in the dataset?
Answer: Directors from IMDb
Status: Proposed
Provenance: directors.json
Relation with other req: Root requirement for directors
Associated ontology Req: Director
- R4;
Competency question:
What producers are represented in the dataset?
Answer: Producers from IMDb

- Status: Proposed
- Provenance: producers.json
- Relation with other req: Root requirement for producers
- Associated ontology Req: Producer
- R5;
- Competency question:
- What is the budget for a given movie?
- Answer: The adjusted amount of money invested in the movie if present
- Status: Proposed
- Provenance: movies.json
- Relation with other req: R1
- Associated ontology Req: hasBudget
- R6;
- Competency question:
- Which actors participated in a given movie?
- Answer:
- One or more persons linked to the movie as actors
- Status: Proposed
- Provenance: movies.json
- Relation with other req: R1,R2,R13,R14
- Associated ontology Req: Movie, Actor, isActedBy
- R7;
- Competency question:
- Which directors participated in a given movie?
- Answer:
- One or more persons linked to the movie as directors
- Status: Proposed
- Provenance: movies.json
- Relation with other req: R1,R3,R16,R18
- Associated ontology Req: Movie, Director, isDirectedBy
- R8;
- Competency question:
- Which producers participated in a given movie?
- Answer:
- One or more persons linked to the movie as producers
- Status: Proposed
- Provenance: movies.json
- Relation with other req: R1,R4,R25,R27
- Associated ontology Req: Movie, Producer, isProducedBy
- R9;
- Competency question:
- What gender percentage does a given movie have?
- Answer:
- Integer percent of female actors in the movie
- Status: Proposed
- Provenance: movies.json
- Relation with other req: R1, R6 (R14)
- Associated ontology Req: Movie, hasGenderPercent

- R10;
Competency question:
What genres does a movie belong to?
Answer:
One or more genres linked to the movie
Status: Proposed
Provenance: movies.json
Relation with other req: R1
Associated ontology Req: Movie, hasGenre
- R11;
Competency question:
What is the release year of a movie?
Answer:
The year property linked to the movie
Status: Proposed
Provenance: movies.json
Relation with other req: R1
Associated ontology Req: Movie, hasReleaseYear
- R12;
Competency question:
What is the title of a given movie?
Answer:
The title linked to the movie
Status: Proposed
Provenance: movies.json
Relation with other req: R1
Associated ontology Req: Movie, hasTitle
- R13;
Competency question:
What is the name of a given actor?
Answer:
The name linked to the actor
Status: Proposed
Provenance: actors.json
Relation with other req: R2
Associated ontology Req: Actor, hasName
- R14;
Competency question:
What is the gender of a given actor?
Answer:
The gender linked to the actor
Status: Proposed
Provenance: actors.json
Relation with other req: R2
Associated ontology Req: Actor, hasGender
- R15;
Competency question:
In what movies did a given actor participate?

- Answer:
One or more movies linked to the actor
Status: Proposed
Provenance: movies.json
Relation with other req: R1, R2
Associated ontology Req: Actor, Movie, actsIn
- R16;
Competency question:
What is the name of a given director?
Answer:
The name linked to the director
Status: Proposed
Provenance: director.json
Relation with other req: R3
Associated ontology Req: Director, hasName
 - R17;
Competency question:
In what movies did a given director participate?
Answer:
One or more movies linked to the director
Status: Proposed
Provenance: directors.json
Relation with other req: R1, R3, R11
Associated ontology Req: Director, Movie, directsA
 - R18;
Competency question:
What is the gender of a given director?
Answer:
The gender linked to the director
Status: Proposed
Provenance: directors.json
Relation with other req: R3
Associated ontology Req: Director, hasGender
 - R19;
Competency question:
Which was the first movie a given director participated in?
Answer:
The year of the first movie directed by the director
Status: Proposed
Provenance: directors.json
Relation with other req: R3
Associated ontology Req: Director, hasFirstMovieYear
 - R20;
Competency question:
Which was the last movie a given director participated in?
Answer:
The year of the last movie directed by the director
Status: Proposed

- Provenance: directors.json
 Relation with other req: R3
 Associated ontology Req: Director, hasLastMovieYear
- R21;
 Competency question:
 What is the male count of a given director?
 Answer:
 The number of male-specific pronouns from director's IMDb bio page
 Status: Proposed
 Provenance: directors.json
 Relation with other req: R3
 Associated ontology Req: Director, hasMaleCount
 - R22;
 Competency question:
 What is the female count of a given director?
 Answer:
 The number of female-specific pronouns from director's IMDb bio page
 Status: Proposed
 Provenance: directors.json
 Relation with other req: R3
 Associated ontology Req: Director, hasFemaleCount
 - R23;
 Competency question:
 Does the given director have actor credits?
 Answer:
 The number of female-specific pronouns from director's IMDb bio page
 Status: Proposed
 Provenance: directors.json
 Relation with other req: R3
 Associated ontology Req: Director, has ActorCredits
 - R24;
 Competency question:
 Does the given director have actress credits?
 Answer:
 A boolean value
 Status: Proposed
 Provenance: directors.json
 Relation with other req: R3
 Associated ontology Req: Director, has ActressCredits
 - R25;
 Competency question:
 What is the name of a given producer?
 Answer:
 The name linked to the producer
 Status: Proposed
 Provenance: producers.json
 Relation with other req: R4
 Associated ontology Req: Producer, hasName

- R26;
Competency question:
In what movies did a given producer participate?
Answer:
One or more movies linked to the producer
Status: Proposed
Provenance: producers.json
Relation with other req: R1, R4,R11
Associated ontology Req: Producer, Movie, producesA
- R27;
Competency question:
What is the gender of a given producer?
Answer:
The gender linked to the producer
Status: Proposed
Provenance: producers.json
Relation with other req: R4
Associated ontology Req: Producer, hasGender
- R28;
Competency question:
Which was the first movie a given producer participated in?
Answer:
The year of the first movie produced by the producer, can be more than one related to the specific role (producer, executive or associate)
Status: Proposed
Provenance: producers.json
Relation with other req: R4
Associated ontology Req: Producer, hasFirstMovieYear, hasFirstProducerMovieYear, hasFirstExecutiveMovieYear, hasFirstAssociateMovieYear
- R29;
Competency question:
Which was the last movie a given producer participated in?
Answer:
The year of the last movie produced by the producer can be more than one related to the specific role (producer, executive or associate)
Status: Proposed
Provenance: producers.json
Relation with other req: R4
Associated ontology Req: Producer, hasLastMovieYear, hasLastProducerMovieYear, hasLastExecutiveMovieYear, hasLastAssociateMovieYear
- R30;
Competency question:
What is the male count of a given producer?
Answer:
The number of male-specific pronouns from producer's IMDb bio page
Status: Proposed

- Provenance: producers.json
- Relation with other req: R4
- Associated ontology Req: Producer, hasMaleCount
- R31;
- Competency question:
- What is the female count of a given producer?
- Answer:
- The number of female-specific pronouns from producer's IMDb bio page
- Status: Proposed
- Provenance: producers.json
- Relation with other req: R4
- Associated ontology Req: Producer, hasFemaleCount
- R32;
- Competency question:
- Does the given producer have actor credits?
- Answer:
- A boolean value
- Status: Proposed
- Provenance: producers.json
- Relation with other req: R4
- Associated ontology Req: Producer, hasActorCredits
- R33;
- Competency question:
- Does the given producer have actress credits?
- Answer:
- A boolean value
- Status: Proposed
- Provenance: producers.json
- Relation with other req: R4
- Associated ontology Req: Producer, hasActoressCredits
- R34;
- Competency question:
- Which movie is rated?
- Answer:
- A movie with ratings
- Status: Proposed
- Provenance: IMDb_ratings.csv
- Relation with other req: R1
- Associated ontology Req: Rating, Movie, hasRating
- R35;
- Competency question:
- What is the average rating of a given movie?
- Answer:
- A numeric value on a 1-10 scale
- Status: Proposed
- Provenance: IMDb_ratings.csv
- Relation with other req: R40
- Associated ontology Req: Rating, hasRatingValue

- R36;
Competency question:
What is the amount of votes of a given rating?
Answer:
A numeric value min 5
Status: Proposed
Provenance: IMDb_ratings.csv
Relation with other req: R40
Associated ontology Req: Rating, hasNumVotes
- R37;
Competency question:
What Academy Award ceremonies are represented in the dataset?
Answer:
The complete list of all Academy Awards Ceremony entities (1st through present) with their unique identifiers and years
Status: Proposed
Provenance: the_oscar_nominees.csv
Relation with other req: R40
Associated ontology Req: Ceremony, hasCeremonyNumber
- R38;
Competency question:
What nominees has an specific ceremony edition?
Answer:
The enumerated set of all Award Category entities; Proposed; field: nominee (not from dataset)
Status: Proposed
Provenance: the_oscar_nominees.csv
Relation with other req: R37
Associated ontology Req: Nominee, Ceremony hasNominee
- R39;
Competency question:
What movies have been nominated for Academy Awards?
Answer:
The comprehensive set of Film entities that received at least one nomination across all ceremonies
Status: Proposed
Provenance: the_oscar_nominees.csv
Relation with other req: R12, R38
Associated ontology Req: Nominee, Movie, hasNominatedMovie
- R40;
Competency question:
What actors, directors, and other professionals have been nominated for Academy Awards?
Answer:
The complete list of Person entities who received at least one nomination in any category
Status: Proposed
Provenance: the_oscar_nominees.csv

- Relation with other req: R13, R16, R26
 Associated ontology Req: Nominee, Actor, Director, Producer, hasNominatedPerson
- R41;
 Competency question:
 What award categories exist in the Academy Awards?
 Answer:
 The enumerated set of all Award Category entities
 Status: Proposed
 Provenance: the_oscar_nominees.csv
 Relation with other req: R38
 Associated ontology Req: Nominee, hasCategory
 - R42;
 Competency question:
 What year was an Academic Award Cerermony celebrated?
 Answer:
 The year of the Oscar ceremony
 Status: Proposed
 Provenance: the_oscar_nominees.csv
 Relation with other req: R37
 Associated ontology Req: Ceremony, hasYearCelebrated
 - R43;
 Competency question:
 Has a nominee won a certain ceremony?
 Answer:
 A boolean value
 Status: Proposed
 Provenance: the_oscar_nominees.csv
 Relation with other req: R38, R40, R41
 Associated ontology Req: Nominee, isWinner

Recursos ontológicos reutilizados

Con tal de favorecer la interoperabilidad y alinearse con los principios FAIR, durante el desarrollo se han reutilizado vocabularios y ontologías ampliamente utilizados en la Web Semántica.

Entre los principales recursos reutilizados destacan:

- **RDF** y **RDFS** para la definición básica de clases y propiedades.
- **OWL** para expresar relaciones semánticas más complejas y restricciones dentro de la ontología.
- **XSD** para definir tipos de datos como enteros, cadenas de texto o valores booleanos.
- **Schema.org**, especialmente conceptos relacionados con películas y personas, como *Movie* o *Person*.

- **FOAF (Friend of a Friend)** para reutilizar propiedades relacionadas con personas, como nombres o identificadores.
- **Dublin Core (DC Terms)** para describir metadatos generales del recurso, como títulos o fechas.
- **SKOS** para representar categorías y conceptos controlados, como los géneros cinematográficos o categorías de premios Óscar.

La reutilización de estos recursos permitió mejorar la compatibilidad de la ontología con otros sistemas y facilitar futuras integraciones con grafos de conocimiento externos.

Herramientas y técnicas de evaluación

Para el desarrollo y evaluación de la ontología se utilizaron distintas herramientas especializadas del ámbito de la Web Semántica.

La ontología fue desarrollada principalmente utilizando CHOWLK y **Protégé**, que permitió modelar clases, propiedades y relaciones de forma visual y validar la estructura general del modelo.

Posteriormente, se utilizaron validadores de sintaxis RDF y OWL para comprobar la consistencia de la ontología y detectar posibles errores semánticos o estructurales. También se generaron y ejecutaron **SHACL Shapes** para verificar restricciones específicas sobre los datos del grafo de conocimiento.

Entre las técnicas de evaluación aplicadas destacan:

- **Validación sintáctica**, para comprobar que los ficheros RDF/TTL estuvieran correctamente definidos.
- **Validación semántica**, verificando coherencia entre clases, dominios y rangos.
- **Validación mediante SHACL**, asegurando que los datos instanciados cumplen las restricciones definidas.
- **Evaluación mediante competency questions**, comprobando que la ontología es capaz de responder a los requisitos planteados inicialmente.

Una vez desarrollada la ontología, utilizamos OOPS! para validarla. Arreglamos un par de recomendaciones menores y una importante (principalmente P10 por faltar axiomas disjuntos). Tras solucionar esos errores, solo nos hemos quedado con 2 menores que hemos decidido ignorar (P04 por elementos inconexos de schema y skos, y P13 por relaciones inversas no declaradas que no hemos visto necesarios) y otro error menor del que somos conscientes (P08 por falta de anotaciones rdfs:comment).

Licencias y nivel de FAIRness

La ontología y el grafo de conocimiento desarrollados siguen los principios FAIR (Findable, Accessible, Interoperable and Reusable) con el objetivo de facilitar su publicación, reutilización e interoperabilidad.

Hemos utilizado FOOPS! para evaluar los principios FAIR de la ontología, obteniendo un 90%. No hemos podido evaluar la accesibilidad, pero del resto de principios solo nos ha

faltado dentro de R1, un 50% de OM3 (faltan metadatos que no veíamos relevantes) y VOC4 (por falta de rdfs:comment como hemos mencionado anteriormente).

TO BE FINDABLE:

- F1. (meta)data are assigned a globally unique and persistent identifier.
- F2. data are described with rich metadata.
- F3. metadata clearly and explicitly include the identifier of the data it describes.
- F4. (meta)data are registered or indexed in a searchable resource

TO BE ACCESSIBLE:

- A1. (meta)data are retrievable by their identifier using a standardized communications protocol.
 - A1.1 the protocol is open, free, and universally implementable.
- A2. metadata are accessible, even when the data are no longer available.

TO BE INTEROPERABLE:

- I1. (meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation.
- I2. (meta)data use vocabularies that follow FAIR principles.

TO BE RE-USABLE:

- R1. meta(data) are richly described with a plurality of accurate and relevant attributes.
 - R1.1. (meta)data are released with a clear and accessible data usage license.
 - R1.2. (meta)data are associated with detailed provenance.

Conclusiones

En este proyecto se ha desarrollado una ontología orientada al dominio cinematográfico a partir de distintos datasets relacionados con películas, actores, directores, productores, valoraciones de IMDb y premios Óscar. El trabajo realizado ha permitido transformar información heterogénea en un modelo semántico estructurado y coherente, capaz de representar relaciones complejas dentro de la industria del cine.

El resultado obtenido es una base sólida para futuras ampliaciones, como la incorporación de nuevos datasets, la conexión con fuentes externas como Wikidata o

DBpedia, o el desarrollo de aplicaciones de análisis y visualización sobre el grafo de conocimiento. Además, el seguimiento de los principios FAIR contribuye a que los recursos desarrollados sean accesibles, reutilizables e interoperables dentro del ecosistema de la Web Semántica.